Formulae

Exam code: 0620 & 0971

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Updated on 30 September 2025

Molecular formulae

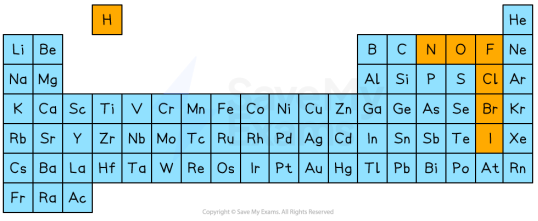
- Elements are often represented using their **chemical symbol** from the Periodic Table

Chemical symbols of elements table

Element	Symbol	Element	Symbol
Lithium	Li	Calcium	Ca
Chromium	Cr	Gold	Au
Aluminium	Al	Sulfur	S

- Seven elements exist as diatomic **molecules**, which means that they are molecules made of two atoms:
  1. Hydrogen, H_2
  2. Nitrogen, N_2
  3. Oxygen, O_2
  4. Fluorine, F_2
  5. Chlorine, Cl_2
  6. Bromine, Br_2
  7. Iodine, I_2

Periodic Table identifying the 7 diatomic elements



The highlighted elements all exist as diatomic molecules with the formula X_2

- These 7 elements are also classed as **simple molecules**
- Atoms combine together in **fixed ratios** that will give them full outer shells of electrons
  - When this happens, a molecule is formed

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

AND

- The number of atoms, given by the subscript (little) number after a chemical symbol

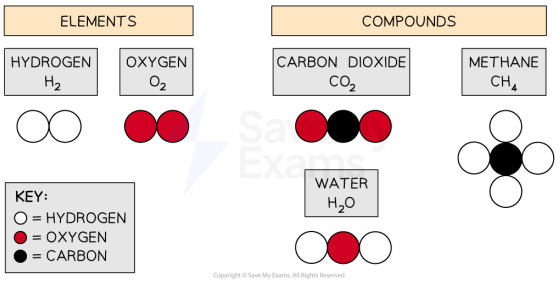
Examples of molecular formulae

Substance	Molecular formula	Made from
Hydrogen	H_2	2 hydrogen atoms
Chlorine	Cl_2	2 chlorine atoms
Water	H_2O	2 hydrogen atoms 1 oxygen atom
Methane	CH_4	1 carbon atom 4 hydrogen atoms
Ammonia	NH_3	1 nitrogen atom 3 hydrogen atoms
Sulfuric acid	H_2SO_4	2 hydrogen atoms 1 sulfur atom 4 oxygen atoms

- The table also shows that the molecular formula can be deduced from the relative number of atoms present
  - E.g. Ammonia contains 3 atoms of hydrogen and 1 atom of nitrogen, which means its molecular formula is NH_3

Diagrammatic representation of chemicals

- Diagrams or models can be used to represent and/or deduce the molecular formula of elements and simple compounds:



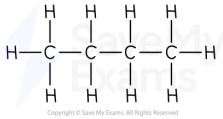
When simple molecules are represented using coloured atoms, a key is used to show the colours for each type of atom

- Using the water examples, it is important to know that these representations also show the arrangement of the atoms in the molecule
  - Water, H_2O

oxygen atom all joined together in a row

Worked Example

What is the molecular formula of the following compound?



Answer:

- The molecule contains:
  - 4 carbon atoms
  - 10 hydrogen atoms
- Therefore, the molecular formula is C_4H_{10}

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